

MedTrack DV — Module 7

QA Checklist | Testing and Validation

Project: MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

Purpose: Validate the implemented KPIs, healthcare operational metrics, patient-flow analytics, and dashboard interactions in the final integrated Tableau workbook.

ID	Validation Area	Test / Check	Expected Result	Observed Result	Status
KPI-01	KPI Calculation	Total Admissions	Calculated from final dataset record count; value stored in final dataset.	100,000 admissions	PASS
KPI-02	KPI Calculation	Occupancy Rate	Calculated from Bed_Occupancy_Rate_Calc and expressed as a percentage.	36.757%	PASS
KPI-03	KPI Calculation	Average Length of Stay	Mean of Length of Stay; expressed in days.	7.50314 days	PASS
KPI-04	KPI Calculation	Readmission Rate	Readmitted patients divided by total admissions × 100.	7.477%	PASS
KPI-05	KPI Calculation	Bed Utilization Rate	Occupied beds divided by total admissions × 100.	50.36%	PASS
KPI-06	KPI Calculation	Department Efficiency Score	Average of bed occupancy, ICU occupancy and staff utilization components.	49.35023%	PASS
KPI-07	KPI Calculation	Staff Utilization Rate	Mean of Staff_Utilization_Calc.	57.92486%	PASS
MET-01	Healthcare Metrics	Admissions metric	Admissions are consistently represented across the KPI and dashboard views.	Total admissions KPI and admission-trend views present	PASS
MET-02	Healthcare Metrics	Length of Stay	Length of Stay is represented in days and used for average LOS analysis.	7.50314 days; LOS dashboard analysis present	PASS
MET-03	Healthcare Metrics	Readmission	Readmission is used for overall and department-level analysis.	7.477% overall; department comparison present	PASS
MET-04	Healthcare Metrics	Bed occupancy/utilization	Bed-related indicators are available for KPI and resource analysis.	Occupancy/utilization views present	PASS
MET-05	Healthcare Metrics	Staff utilization	Staff utilization is implemented as an additional operational KPI.	57.92486%; Staff Utilization by Department present	PASS
MET-06	Healthcare Metrics	Department efficiency	Department efficiency is available for department comparison.	Department Efficiency Score view present	PASS
HO-01	Hospital Overview	Hospital performance KPIs	Four dashboard KPI cards display the required overview measures.	KPI cards displayed	PASS
HO-02	Hospital Overview	Patients by Department	Department patient volume is displayed as a comparison chart.	Chart present	PASS

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HO-03	Hospital Overview	Patient distribution by hospital type	Hospital-type patient distribution is displayed.	Chart present	PASS
HO-04	Hospital Overview	Top states by patient volume	State-level patient volume is displayed.	Chart present	PASS
HO-05	Hospital Overview	Monthly patient load trend	Monthly patient admissions/load are displayed as a trend.	Line chart present	PASS
PF-01	Patient Flow	Admission analysis	Admission-related patient volume is displayed by admission type and over time.	Admission views present	PASS
PF-02	Patient Flow	Discharge tracking	Monthly discharge trend is displayed.	Line chart present	PASS
PF-03	Patient Flow	Patient movement	Transfers by department are displayed.	Transfer chart present	PASS
PF-04	Patient Flow	Average stay analysis	Average length of stay is represented through KPI/analysis.	LOS KPI/analysis present	PASS
PF-05	Patient Flow	Patient load monitoring	Monthly patient load trend is displayed.	Trend chart present	PASS
DA-01	Department Analytics	Patient volume comparison	Patient volume is compared across departments.	Chart present	PASS
DA-02	Department Analytics	Readmission by department	Readmission is compared by department.	Chart present	PASS
DA-03	Department Analytics	Average LOS by department	Average LOS is compared across departments.	Chart present	PASS
DA-04	Department Analytics	Department efficiency	Department Efficiency Score is displayed for comparison.	Chart present	PASS
DA-05	Department Analytics	Treatment capacity	Treatment capacity is analyzed.	Chart present	PASS
RU-01	Resource Utilization	Staff utilization	Staff utilization is compared by department.	Chart present	PASS
RU-02	Resource Utilization	Equipment utilization	Equipment utilization is displayed.	Chart present	PASS
RU-03	Resource Utilization	Resource availability	Resource availability is analyzed.	Chart present	PASS
RU-04	Resource Utilization	Capacity planning	Capacity planning insights are displayed.	Chart present	PASS
RU-05	Resource Utilization	Bed occupancy by hospital type	Bed occupancy is compared by hospital type.	Chart present	PASS
INT-01	Dashboard Interaction	State filter	Selecting a state filters the dashboard analysis.	State filter implemented	PASS
INT-02	Dashboard Interaction	Department filter	Selecting a department filters the dashboard analysis.	Department filter implemented	PASS
INT-03	Dashboard Interaction	Cross-filtering	Chart selections update related dashboard views where configured.	Interactive filtering implemented	PASS
INT-04	Dashboard Interaction	Department selection parameter	Department selection can be used for the selected-department display/analysis.	Parameter action implemented	PASS

ID	Validation Area	Test / Check	Expected Result	Observed Result	Status
INT-05	Dashboard Interaction	Dashboard navigation	Navigation controls switch between the four dashboards.	Navigation buttons/actions implemented	PASS
INT-06	Dashboard Interaction	Dashboard integration	All four dashboards are contained in the final integrated workbook.	Four dashboards present in MedTrack_DV.twbx	PASS

QA Summary: All listed validation checks passed. No major dashboard issues were identified in the documented validation scope.

KPI validation note: The seven implemented KPI values were reproduced from the final dataset using the implemented KPI definitions. This establishes formula-level/numerical reproducibility. It should not be presented as an independently measured statistical accuracy percentage against an external ground-truth dataset.